Sleep Medicine 131 (2025) 106485



Contents lists available at ScienceDirect

Sleep Medicine

journal homepage: www.elsevier.com/locate/sleep





Separating obstructive and central respiratory events during sleep using breathing sounds: Utilizing transfer learning on deep convolutional networks

Shumit Saha [a,b,c,d], Nasim Montazeri Ghahjaverestan [c,e], Azadeh Yadollahi [b,c,*]

[a] Department of Biomedical Data Science, School of Applied Computational Sciences, Meharry Medical College, Nashville, TN, USA
[b] Institute of Biomedical Engineering, University of Toronto, Toronto, ON, Canada
[c] KITE-Toronto Rehabilitation Institute, University Health Network, Toronto, ON, Canada
[d] Institute of Health Policy, Management, and Evaluation, Dalla Lana School of Public Health, University of Toronto, Toronto, ON, Canada
[e] Department of Electrical and Computer Engineering, Queens University, London, ON, Canada

A B S T R A C T

Sleep apnea diagnosis relies on polysomnography (PSG), which is resource-intensive and requires manual analysis to differentiate obstructive sleep apnea (OSA) from central sleep apnea (CSA). Existing portable devices, while valuable in detecting sleep apnea, often do not distinguish between the two types of apnea. Such differentiation is critical because OSA and CSA have distinct underlying causes and treatment approaches. This study addresses this gap by leveraging tracheal breathing sounds as a non-invasive and cost-effective method to classify central and obstructive events.

We employed a transfer learning strategy on six pre-trained deep convolutional neural networks (CNNs), including Alexnet, Resnet18, Resnet50, Densenet161, VGG16, and VGG19. These networks were fine-tuned using spectrograms of tracheal sound signals recorded during PSG. The dataset, comprising 50 participants with a combination of central and obstructive events, was used to train and validate the model. Results showed high accuracy in differentiating central from obstructive respiratory events, with the combined CNN architecture achieving an overall accuracy of 83.66 % and a sensitivity and specificity above 83 %.

The findings suggest that tracheal breathing sounds can effectively distinguish between OSA and CSA, providing a less invasive and more accessible alternative to traditional PSG. This methodology could be implemented in portable devices to enhance the diagnosis of sleep apnea, enabling targeted treatment. By facilitating earlier and more accurate diagnoses, this method supports personalized treatment strategies, optimizing therapy selection (e.g., CPAP for OSA, ASV for CSA) and ultimately enhancing clinical outcomes.

1. Introduction

Sleep apnea is a chronic respiratory disorder that occurs due to the recurrent complete or partial cessations of breathing during sleep [1]. The assessment of sleep apnea severity is determined by the apnea-hypopnea index (AHI), which shows the number of apneas (full cessation of breathing) and hypopneas (partial cessation of breathing) per hour of sleep. The gold standard assessment of sleep apnea is polysomnography (PSG), which requires the participant to sleep with around 20 sensors attached to their body. The complex nature of PSG, its high cost, and long waiting time have led to a growing application of portable devices to diagnose sleep apnea [2]. One challenge is that, most of the portable devices have focused on estimating the AHI without distinguishing the types of sleep apnea. The type of sleep apnea is divided into obstructive sleep apnea (OSA) and central sleep apnea (CSA). While OSA is characterized by collapse and obstruction in the pharyngeal airway, CSA is caused by the loss of central neural respiratory drive to the respiratory muscles [3]. The prevalence of OSA and CSA in the adult population is 10 % [4] and 0.9 % [5], respectively. Although OSA is far more common in the general population than CSA, CSA is highly prevalent in patients with cardiovascular disorders such as heart failure (40 %) or stroke (7–24 %) [6–8]. Moreover, the most common treatment of OSA is continuous airway positive pressure (CPAP), while CSA is treated with adaptive servo-ventilation or by managing the underlying comorbidities [3]. However, CPAP treatment could increase death and morbidity in patients with CSA [6,9]. Therefore, a complete characterization of CSA and OSA is required to administer the effective treatment of sleep apnea. Thus, it is important to develop algorithms to distinguish between central and obstructive respiratory events using portable monitoring of sleep apnea.

The gold standard for accurate differentiation between central and obstructive respiratory events is the esophageal manometry [10]. In this

* Corresponding author. Toronto Rehabilitation Instititue, University Health Network, Toronto, ON, M5G 2A2, Canada
E-mail address: Azadeh.Yadollahi@uhn.ca (A. Yadollahi).

https://doi.org/10.1016/j.sleep.2025.106485
Received 9 September 2024; Received in revised form 25 February 2025; Accepted 28 March 2025
Available online 29 March 2025
1389-9457/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

method, an esophageal catheter is inserted through the nose into the esophagus to monitor the respiratory effort. While the respiratory effort is present during obstructive events, there is no or low respiratory effort during central events [10]. However, the measurement of esophageal pressure is invasive and inconvenient and cannot be performed during clinical PSG. Therefore, the American Academy of Sleep Medicine (AASM) has recommended to discriminate OSA from CSA using clinical PSG data by manual inspection of the chest and abdominal movement along with nasal airflow. In comparison to esophageal manometry, manual inspection of the respiratory movement and nasal airflow from PSG can distinguish OSA from CSA with 95 % accuracy [10].

Most portable devices for sleep apnea diagnosis record nasal airflow [11,12], peripheral blood oxygen saturation level (SaO_2), tracheal sound [13–16], or chest and abdominal movements [17–19]. Although a large body of studies has focused on estimating the AHI and diagnosing sleep apnea, very few studies have focused on distinguishing between obstructive and central respiratory events or diagnosing patients with dominant CSA [20–24]. Thomas et al. have used the electrocardiogram (ECG) and showed that the spectral analysis of ECG could separate the patients with dominant OSA from CSA [23]. Pillar et al. have used accelerometer-based respiratory movements for diagnosing CSA with 67 % sensitivity and 100 % specificity [21]. Sommermeyer et al. developed an automatic algorithm using nasal airflow and pulse oximeter; and estimated the central apnea index with a correlation coefficient of 0.87 compared with PSG [22]. A recent study by Akbarian et al. has demonstrated the application of an infrared video camera to distinguish between obstructive and central apnea [24]. However, these studies have not distinguished whether each respiratory event is obstructive or central, neither they classified hypopneas into obstructive or central events. Hypopneas are more common than apneas and it is important to categorize hypopneas for better assessment of sleep apnea [25]. Furthermore, detecting central hypopneas are more challenging than apneas, as clinical PSG does not measure the respiratory effort but assesses some of the consequence of central hypopneas. Several clinical observations have used airflow, chest, and abdominal movements to detect the central hypopneas manually [26–28]. However, distinguishing between central and obstructive hypopneas using an automatic algorithm has not been readily investigated.

Recently, breathing sound analysis has been shown to be a strong modality in managing sleep apnea. Breathing sounds can be recorded using a microphone in wearable devices, providing a convenient and non-intrusive monitoring method. Analysis of these sounds helps quantify pharyngeal airway narrowing and pinpoint the site of pharyngeal collapse. [29–32]. Furthermore, acoustic sounds, such as breathing, snoring, voice-based technologies have been extensively used to monitor respiration during sleep, estimate the sleep apnea severity, and understanding the underlying factors of upper airway narrowing [13,14,29,33–44]. Moreover, breathing sounds analysis has shown promising results in assessing the treatment of sleep apnea [45–47]. However, breathing sounds-based separation of central and obstructive events has not been readily implemented. Previously, Alshaer et al. performed a pilot study on 85 events and demonstrated the application of breathing sounds on separating central and obstructive events [33]. Thus, a study is required to evaluate the application of breathing sounds to distinguish the obstructive from central events in larger samples. In this study, we aim to develop an automatic machine-learning algorithm to differentiate between obstructive and central respiratory events using tracheal breathing sounds.

2. Methods

The study received approval from the Research Ethics Board (REB) of the University Health Network (UHN) (Approval Number: 15–8967) ensuring compliance with ethical guidelines for human research. Prior to participation, all individuals provided written informed consent, acknowledging their understanding of the study procedures, potential risks, and their right to withdraw at any time.

2.1. Study participants

We recruited the individuals referred for diagnosis of sleep apnea in the sleep laboratory of the University Health Network-Toronto Rehabilitations Institute. Eligible participants were adults aged 18–85 years, with no restrictions based on sex or body mass index. The study excluded individuals with allergies to clinical tape and those with insufficient sleep data, defined as total sleep time of less than 2 h [14]. In this study, we excluded the individuals who did not have any central or obstructive respiratory events during sleep.

2.2. Data measurement

2.2.1. Sleep study

We performed the overnight Polysomnography (PSG) using Embla® N7000/S4500 (Natus Medical Incorporated). Thoracoabdominal movement, nasal pressure, and SaO_2 were monitored using respiratory inductance plethysmography (Ambulatory Monitoring Inc.), nasal cannulae, and oximetry (Nonin Xpod® 3012LP), respectively. Furthermore, PSG monitored the brain activity by electroencephalography (EEG), heart rhythm by electrocardiogram (ECG), eye movement by electrooculogram (EOG), and chin and leg movement by electromyogram (EMG).

The central and obstructive events were annotated based on the thoracoabdominal movement and nasal pressure, following the American Academy of Sleep Medicine (AASM) guidelines [1]. Obstructive apnea was defined based on the presence of respiratory effort (assessed by thoracoabdominal movement) while the nasal airflow was absent. Central apnea was defined when there was no respiratory effort while the nasal airflow was absent. Hypopnea was defined as $\geq$30 % decrease in breathing airflow for more than 10 s in association with blood oxygen desaturation of $\geq$3 % or arousal. Obstructive hypopnea was defined when the increased inspiratory flattening of the nasal pressure in association with the thoracoabdominal paradox was observed, or there was a presence of snoring. If all the criteria did not meet, then the hypopnea was termed as central. The events were annotated by sleep technologists with more than 10 years of experience. The events were further annotated by another researcher who was blind to the sleep technician's assessment. We achieved an agreement of 87 % in detecting events with the sleep technician. The discrepant events were excluded from the analysis.

2.2.2. Recording of breathing sounds

We used a wearable device called the Patch, which was developed by our group to record breathing sounds [14]. The Patch consists of a microphone and a 3-dimensional accelerometer. We attached the Patch to the suprasternal notch using double-sided adhesive tape. The sound signal was digitized and saved on an internal memory using sampling rates of 15,300 Hz. Since breathing sounds typically range up to 5000 Hz, this rate ensures proper signal representation based on the Nyquist theorem while preserving key airflow and obstruction-related features. A microcontroller embedded in the Patch was programmed to synchronize the Patch with the PSG system. Details of the Patch can be found in our previous studies [39,40].

2.3. Data analysis

2.3.1. Preprocessing

The recorded respiratory sounds were bandpass filtered between 200 and 3000 Hz to remove heart sounds and high-frequency noises [41]. We calculated the spectrogram of the filtered sound signals. To estimate the spectrogram, we used short-time Fourier Transforms (STFT) with a Tuckey window of 20 ms and 90 % overlap between adjacent windows. Spectrograms are 2D images representing the sequence of spectra with

time in the horizontal axis and frequency in the vertical axis, and the color of the spectrogram represents the strength of the frequency component at each time frame.

Figs. 1 and 2 show examples of central and obstructive respiratory apnea and hypopnea, 5-s before and after the events, along with their corresponding spectrograms. While no visible breathings were found during central events (Fig. 1), breathings with smaller amplitudes compared to before and after events were observed during obstructive events (Fig. 2). Furthermore, some obstructive hypopneas were associated with snoring during the events. As respiratory events span from 10 s to more than 50 s, it is crucial to make a fixed dimension of the spectrogram image to avoid the discrepancy created by the duration of the events. Therefore, we selected the 5 s before the events and 10 s within the events irrespective of the event duration and created the 2D spectrogram of 15 s (shown in the red box in Figs. 1 and 2).

2.3.2. *Transfer learning*

In this study, we applied the transfer learning strategy. The transfer learning approach consists of using the convolutional layers of deep network architecture as feature extractors and adjusting only the fully connected layers to the new problem (48). Transfer learning has proven to be an effective strategy for small datasets and gained immediate attention in the biomedical image analysis [48]. In this study, we applied a transfer learning strategy to adapt six deep convolutional neural network (CNN) architectures trained for visual object recognition. The considered architectures were Alexnet [49], Resnet18 [50], Resnet50 [50], VGG16, VGG19 [51], Densenet161 [52]. We selected these networks as the number of parameters in these networks varied from a few thousand to millions which would help us to identify the optimum number of parameters for solving our problem. Furthermore, these network weights were available to be modified.

Fig. 3 shows our approach to utilize transfer learning in distinguishing the central and obstructive events. The 15 s spectrogram images were first padded to the size of 224 × 224. The spectrogram images were then fed into the pre-trained models. The pre-trained models were downloaded from the "torchvision" and loaded onto corresponding architecture [53]. The convolutional layers were fixed to avoid modification on the training phase, and the fully connected layers were modified. Table 1 shows the details of the architectures transferred to our domain. In all architecture, network parameters were optimized using the cross-entropy loss function with Adam optimizer. As obstructive events were more common than central events, we implemented a class weight of 1.4 for central events and 0.2 for obstructive events. We used 20 epochs for training the network. The whole training set was passed to the model in mini-batches of 32 elements.

Finally, we combined the results of six networks and averaged the outputs (Fig. 4). If the averaged output was greater than 0.7, then we marked that image as an obstructive respiratory event. Otherwise, we marked the event as a central respiratory event. The selection of 0.7 was determined by a threefold cross validation technique in the training dataset.

The training and testing were performed under the operating environment where Intel® Core™ i5-7400 CPU @ 3.00 GHz with 8 GB RAM and NVIDIA GeForce GTX 1070 graphic processing unit card with a graphics memory of 8 GB was installed. The Pytorch library was used to implement the algorithm.

2.3.3. *Validation and evaluation*

At first, the dataset was divided into test and remaining sets. The remaining dataset was divided into training and validation datasets. We selected 90 % of the remaining dataset as a training dataset and 10 % as a validation dataset. We evaluated the performance of the architectures

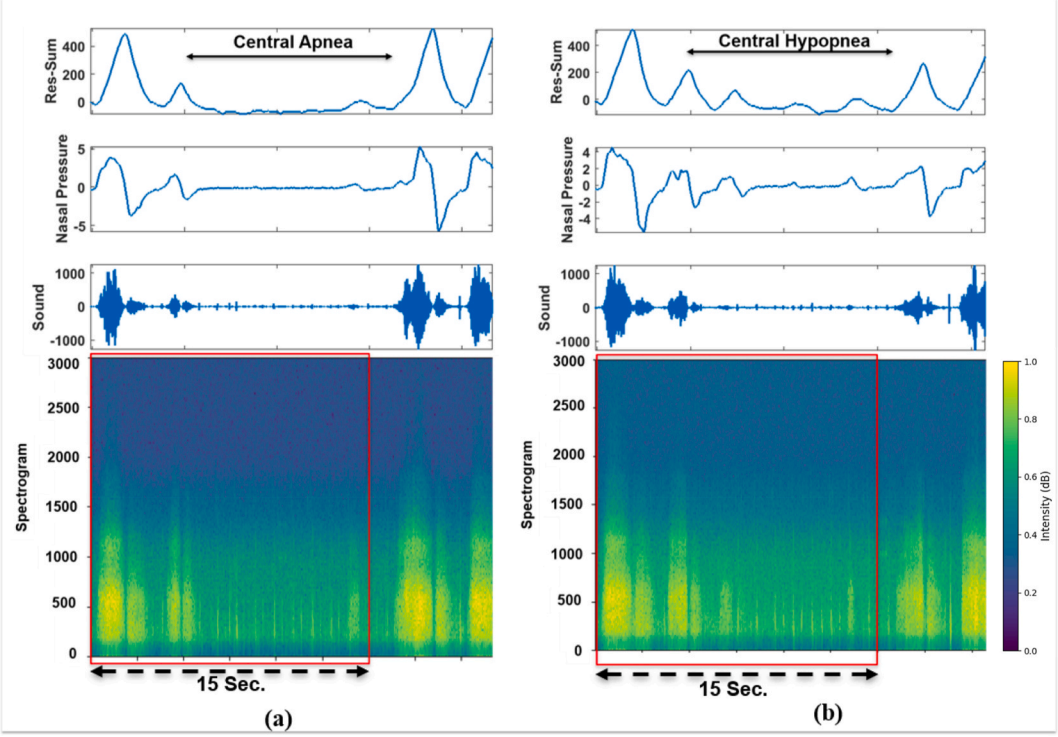


Fig. 1. Central respiratory event example for a) apnea and b) hypopnea. First and second graphs show the thorachoabonimal movement (Res-Sum) and nasal airflow from polysomnography (PSG), respectively. The third and fourth graphs show the filtered breathing sounds and extracted spectrograms, respectively. The color shading represents sound intensity, measured in decibels (dB), and both images are standardized with the same intensity range, making the color bar applicable to both images for direct comparison. We find no visible breathing during central events. Here, red box marks a fixed 15-s segment used for spectrogram analysis, ensuring a consistent comparison across events of varying durations, some of which may exceed 50 s. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

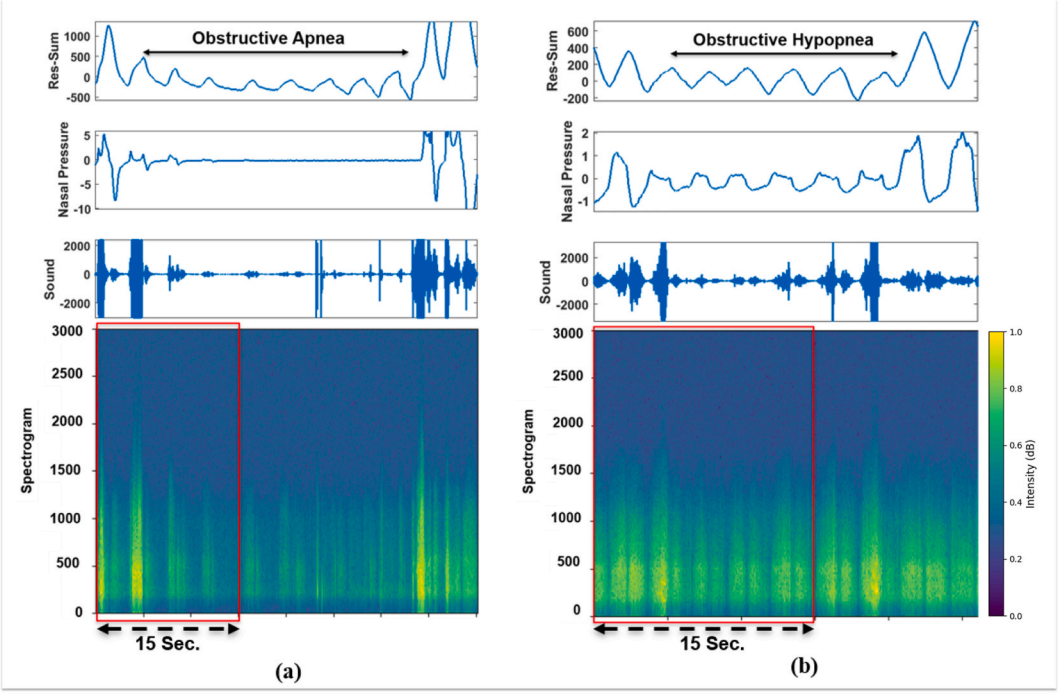


Fig. 2. Obstructive respiratory event example for a) apnea and b) hypopnea. The first and second graphs show the thorachoabonimal movement (Res-Sum) and nasal airflow from polysomnography (PSG), respectively. The third and fourth graphs show the filtered breathing sounds and extracted spectrograms, respectively. The color shading represents sound intensity, measured in decibels (dB), and both images are standardized with the same intensity range, making the color bar applicable to both images for direct comparison. We find shallow amplitude breathing during obstructive events. Here, red box marks a fixed 15-s segment used for spectrogram analysis, ensuring a consistent comparison across events of varying durations, some of which may exceed 50 s. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

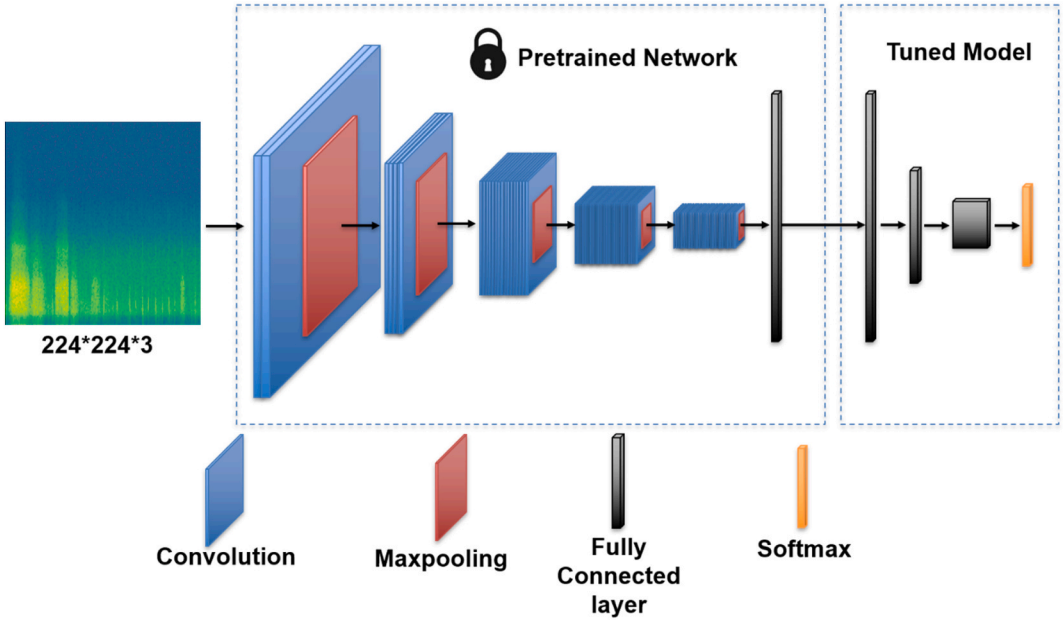


Fig. 3. Proposed transfer learning approach for distinguishing between central and obstructive events. Here, pre-trained networks are Alexnet, Resnet18, Resnet50, Densenet161, VGG16, and VGG19.

in the testing set and calculated the sensitivity, specificity, and accuracy. Here, sensitivity and specificity denoted the success rate of detecting the obstructive and central event detection, respectively. We further calculated the area under the receiver operating curve (ROC-AUC) and selected the best architectures based on the highest ROC-AUC. Furthermore, we calculated the precision, recall, and F1 score to assess the performance of the architectures. Additionally, we reported the performance of the architectures for detecting central and obstructive events for apneas and hypopneas, separately.

Table 1
Details of the different architectures transferred to our domain.

Network	Total Parameters	Training Parameters
Alexnet	4,829,762	2,360,066
Resnet18	11,228,014	51,502
Resnet50	24,033,090	525,058
Densenet161	27,038,018	566,018
VGG16	135,309,890	1,049,346
VGG19	140,619,586	1,049,346

3. Results

We recruited 85 participants for this study. Among them, data of 16 participants were excluded from the analysis due to low sound signal quality (signal-to-noise ratio less than 20) or issues with synchronization between the Patch and PSG. Nineteen subjects did not have any central events during sleep; hence we excluded those subjects. In total, we had 50 subjects. Table 2 shows the demographics of the participants. We found a total number of 661 central and 3780 obstructive respiratory events. Among them, we selected 251 central and 410 obstructive respiratory events spectrograms as the test set images. The rest of the 410 central and 3370 obstructive events were selected for the training and validation set.

Table 3 shows the performance of each convolutional network along with when their combined architecture. Although, the sensitivity of Alexnet (94.39 %), Resnet18 (86.10 %), Resnet50 (91.46 %), Densenet161 (91.22 %) was high, their specificity (Alexnet: 60.16 %; Resnet18: 74.90 %; Resnet50: 63.75 %; Densenet161: 64.94 %) was notably low, which indicated that these networks often misclassified the central respiratory events. Denser networks such as VGG16 and VGG19 maintained high sensitivity (VGG16: 83.66 %; VGG19: 86.34 %) and specificity (VGG16: 77.29 %; VGG19: 76.5 %).

As assessed by the ROC-AUC in Fig. 5-a, we observed the best performance when we combined the results of the six architectures. The results from combining all six architectures showed that the sensitivity and specificity were above 83 %, along with ROC-AUC of 0.84. We also found that the precision, recall, and F1-score of the combined network were 89.12 %, 83.9 %, and 86.43 %, respectively. Fig. 5-b shows the confusion matrix of the detected and true respiratory events for the combined network in the test dataset.

Table 4 shows the performance of the combined network for detecting central and obstructive apneas and hypopneas. The overall accuracy of distinguishing obstructive and central apneas was 87 %, while the accuracy of detecting obstructive and central hypopneas was 83 %. The performance of detecting obstructive apnea (75 %) was lower than in central apnea (89.77 %). In the case of hypopneas, the performance of the combined network was higher for obstructive (84.36 %) events than central ones (79.75 %).

4. Discussion

The most important finding of our study was that tracheal breathing sounds could distinguish the central vs. obstructive respiratory events with high accuracy.

Properly distinguishing between CSA and OSA is important for the selection of appropriate treatment, reducing the risk of comorbidities, and improving the patient's quality of life. CSA is associated with a higher risk of heart failure and stroke, and a higher rate of hospitalization than those with OSA [54,55]. Additionally, CSA is also associated with a higher risk of opioid use. patients who use opioids are at a higher risk of developing CSA, due to the respiratory depressant effect of opioids, which can lead to serious consequences such as hypoxia, respiratory failure, and even death [56,57]. Moreover, the two types of sleep apnea have different pathophysiology and therefore require different treatment approaches. Treatment options for CSA include medications such as acetazolamide and theophylline [58,59], which can help improve the brain's ability to signal the muscles that control breathing. Thus, proper diagnosis of CSA and OSA is crucial and any portable diagnosis of sleep apnea without the distinction between CSA and OSA is not complete.

In recent years, acoustic analysis of breathing sounds for portable diagnosis of sleep apnea has gained rapid attention [60–62] because it is convenient and cost-effective to acquire breathing sounds with a microphone. Due to such convenience, some studies went beyond sleep apnea diagnosis and expanded acoustic analysis in understanding underlying sleep apnea pathophysiology. Sleep apnea is characterized by upper airway obstruction and results in a reduction of nasal airflow. Thus, several studies have also expanded on breathing sounds-based

Table 2
Participant's characteristics information.

Characteristics	Mean ± STD
N (Male/Female)	50 (29/21)
Body mass index (BMI), kg/m^2	29.6 ± 5.6
Age, year	53.06 ± 14.8
Sleep efficiency, %	74.7 ± 17.4
Total sleep time, hr	5.3 ± 1.1
Arousal index, events/hr	29.4 ± 19.4
Sleep stage 1, %	13.8 ± 9.5
Sleep stage 2, %	66.1 ± 9.5
Sleep stage 3, %	6.0 ± 8.0
REM sleep stage, %	14.2 ± 7.0
Mean SaO_2 during sleep, %	94.1 ± 2.5
Minimum SaO_2 during sleep, %	85.3 ± 7.2
AHI, events/hr	20.7 ± 21.9
OAHI, events/hr	16.6 ± 19.8
CAHI, events/hr	2.9 ± 6.8

*AHI: Apnea-hypopnea index, OAHI: Obstructive AHI, CAHI: Central AHI, REM: Rapid eye movement, SaO_2: Blood oxygen saturation.

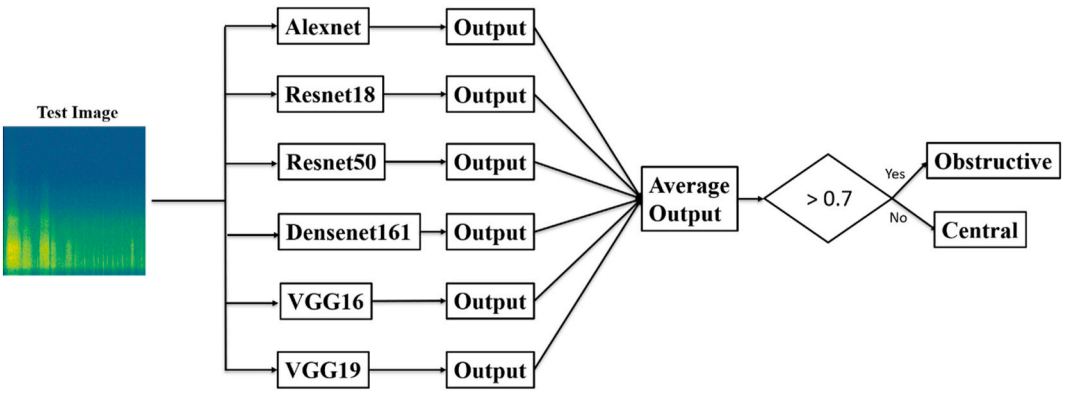


Fig. 4. The combination of the 6 networks by averaging the outputs.

Table 3
Results of obstructive and central respiratory events detection.

Network	Sensitivity	Specificity	Accuracy	ROC-AUC	Precision	Recall	F1
Alexnet	94.39	60.16	81.39	0.77	79.47	94.39	86.29
Resnet18	86.1	74.9	81.85	0.80	84.86	86.1	85.48
Resnet50	91.46	63.75	80.94	0.78	80.47	91.46	85.61
Densenet161	91.22	64.94	81.24	0.78	80.95	91.22	85.78
VGG16	83.66	77.29	81.24	0.81	85.75	83.66	84.69
VGG19	86.34	74.5	81.85	0.80	84.69	86.34	85.51
Combined	**83.9**	**83.27**	**83.66**	**0.84**	**89.12**	**83.9**	**86.43**

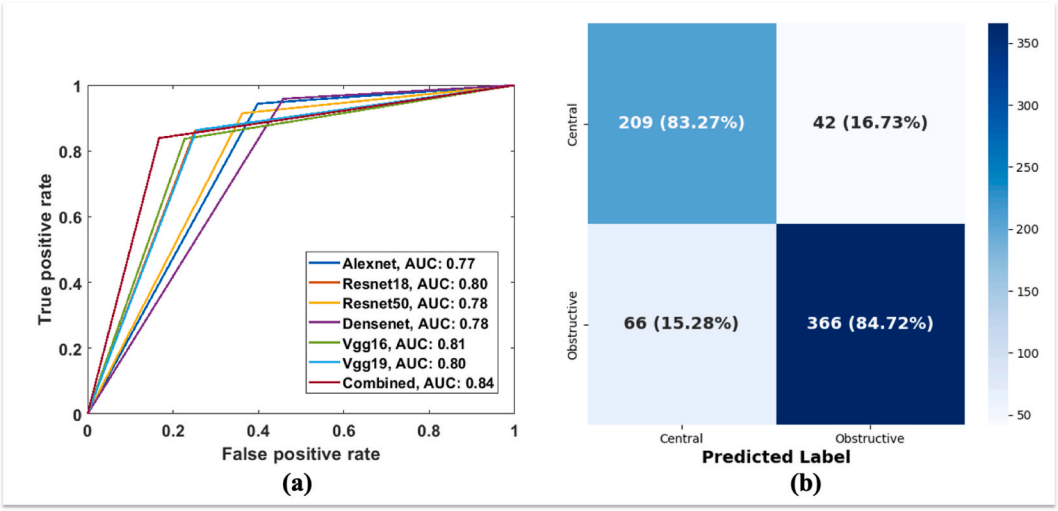


Fig. 5. a) Receiver operating curve (ROC) of classifying central and obstructive respiratory events using the networks. b) the confusion matrix of detecting obstructive and central events in the testing dataset using the combined network.

Table 4
Results of obstructive and central respiratory events detection during Apnea and Hypopnea.

Events	Sensitivity	Specificity	Accuracy	ROC-AUC	Precision	Recall	F1
Apnea	75	89.77	87.04	0.82	62.5	75	68.18
Hypopnea	84.36	79.75	83	0.82	90.88	84.36	87.5

prediction of the nasal airflow [63,64]. Furthermore, sleep apnea can occur in several sites of the upper airway, such as behind the soft palate or at the base of the tongue. Several studies have shown that breathing sounds have great potential in assessing the site of upper airway obstruction [29,30,32,41]. Taken together, these studies have shown the utility of breathing sounds in determining the physiology of the upper airway along with the diagnosis of sleep apnea. However, the distinguishing between central and obstructive respiratory events through breathing sounds has not been readily investigated.

The frequency and magnitude of the breathing sounds could be different between obstructive and central events. Obstructive events are associated with narrowing in the upper airway [65]. Obstructive respiratory events are characterized by pharyngeal airway narrowing due to negative intrathoracic pressure, increased soft tissue volume, and reduced pharyngeal dilator muscle activity. During obstructive apnea, complete airway collapse limits airflow, resulting in low-amplitude breathing sounds. In obstructive hypopnea, partial collapse increases airflow turbulence, leading to pharyngeal tissue vibration and snoring sounds [66]. On the contrary, upper airway narrowing is not a prerequisite for central events. As the caliber of the upper airway may be wider during the central than obstructive events, the turbulence of the airflow is expected to be lower. Thus, lower turbulence can result in low-amplitude breathing sounds. Overall, central events may result in lesser turbulent and lower breathing sound energy than obstructive events.

To capture such subtle changes in turbulence and amplitude, we performed the spectrogram analysis. Spectrogram gives the joint information from both the time and frequency domains. While the time-domain function demonstrates how the amplitude of the sound signal changes over time, the frequency domain function indicates how often such changes take place [67]. As the occurrence of respiratory events can result in changes in energy and turbulence of breathing sounds over time, analyzing breathing sounds in the time-frequency domain can reveal detailed information. Furthermore, spectrogram analysis transferred the breathing sound signal into images which is more suitable for the CNN-based approaches. Our results suggested that spectrogram analysis of breathing sounds has shown great promise in separating respiratory events. Therefore, future studies can consider the time-frequency analysis of breathing sounds in sleep apnea diagnosis.

This study utilized the transfer learning approach to distinguish between obstructive and central events during sleep. The transfer learning approach is a state-of-the-art technique that uses deep CNN architectures and has proven to be successful in biomedical image analysis [68–71]. The success of transfer learning in biomedical imaging is due to the fact that deep CNN architectures are a well-trained network for image classification. Consequently, using a pre-trained network also does not require a big dataset for fine-tuning the model. Utilizing the transfer learning approach in six well-known CNN architectures, we

found more than 80 % accuracies in classifying the central and obstructive events. Furthermore, combining the results of these six CNN architectures showed the best result. Taken together, our results demonstrated the fact that it was not necessary to train a model from scratch to achieve high accuracy while analyzing the breathing sounds. Pre-trained models can be adapted and tuned to achieve high accuracy in sleep apnea-type classification using breathing sounds.

The findings of this study have significant clinical implications, particularly in improving the diagnosis and management of sleep apnea. Traditional PSG remains the gold standard for distinguishing CSA and OSA, but it is resource-intensive, requiring overnight monitoring in a sleep lab. Our results demonstrate that tracheal breathing sounds can accurately differentiate CSA from OSA, enabling the development of portable, non-invasive diagnostic tools that could be used in home sleep monitoring. This research is particularly relevant for underserved populations and resource-limited settings, where access to PSG is restricted. By incorporating this classification algorithm into wearable or bedside monitoring systems, clinicians could more efficiently screen and diagnose sleep apnea subtypes, leading to earlier intervention and personalized treatment. Furthermore, precise classification of CSA and OSA is crucial for guiding therapy selection, such as continuous positive airway pressure (CPAP) for OSA versus adaptive servo-ventilation (ASV) for CSA. Ultimately, this research contributes to enhancing sleep medicine by improving diagnostic accessibility, reducing healthcare costs, and optimizing patient outcomes.

Our study was subject to some limitations. Our proposed method was developed based on a study performed in a sleep laboratory. In a home setting, the algorithm may encounter different types of noises, which would limit the application of our developed algorithms. Therefore, future studies should validate the algorithm in home settings. Furthermore, we developed our algorithm based on the central and obstructive events extracted from clinical PSG, rather than the gold-standard esophageal manometry. Although two observers annotated the data and all discrepant data have been removed from the analysis, we acknowledge that manual determination of events may be susceptible to subjective errors. A future study with esophageal manometry is required for a more robust estimation of the events. Lastly, we had a limited sample size for central events. Future studies are required to validate the current algorithm with a larger sample size.

In conclusion, we showed that breathing sounds could be used to distinguish between central and obstructive respiratory events. Our proposed method could be easily implemented in a wearable device that uses a microphone and can be attached to the suprasternal notch. Our proposed approach will help to improve the diagnosis of sleep apnea by providing the types of sleep apnea. Finally, implementing our model could help physicians to determine the optimum treatment by identifying patients with dominant obstructive or central sleep apnea.

CRediT authorship contribution statement

Shumit Saha: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Nasim Montazeri Ghahjaverestan:** Writing – review & editing, Methodology, Data curation. **Azadeh Yadollahi:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used “ChatGPT 4o” in order to refine grammar and improve stylistic clarity. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Funding sources

This research was funded by operating grants from NSERC (RGPIN-2016-06549).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

SS was supported by Connaught International Scholarship for Doctoral Student.

References

[1] Berry RB, Brooks R, Gamaldo C, Harding SM, Lloyd RM, Quan SF, et al. AASM scoring manual updates for 2017 (version 2.4). J Clin Sleep Med 2017;13(5):665–6.
[2] Corral-Peñafiel J, Pepin J-L, Barbe Ferran. Ambulatory monitoring in the diagnosis and management of obstructive sleep apnoea syndrome. %J European Respiratory Review 2013;22(129):312–24.
[3] Dempsey JA, Veasey SC, Morgan BJ, O'Donnell CP. Pathophysiology of sleep apnea. Physiol Rev 2010 Jan 1;90(1):47–112.
[4] Peppard PE, Young T, Barnet JH, Palta M, Hagen EW, Hla KM. Increased prevalence of sleep-disordered breathing in adults. Am J Epidemiol 2013;177(9):1006–14.
[5] Donovan LM, Kapur VK%JS. Prevalence and characteristics of central compared to obstructive sleep apnea: analyses from the. Sleep Heart Health Study Cohort 2016;39(7):1353–9.
[6] Javaheri S. Sleep disorders in systolic heart failure: a prospective study of 100 male patients. The final report. Int J Cardiol 2006;106(1):21–8.
[7] Wolk R, Kara T, Somers VK. Sleep-disordered breathing and cardiovascular disease. J Circulation 2003;108(1):9–12.
[8] Lyons OD, Ryan CM. Sleep apnea and stroke. Can J Cardiol 2015;31(7):918–27.
[9] Javaheri S. CPAP should not be used for central sleep apnea in congestive heart failure patients. J Clin Sleep Med 2006;2(4):399–402.
[10] Argod J, Pepin J-L, Levy P. Differentiating obstructive and central sleep respiratory events through pulse transit time. Am J Respir Crit Care Med 1998;158(6):1778–83.
[11] Grover SS, Pittman SD. Automated detection of sleep disordered breathing using a nasal pressure monitoring device. Sleep Breath 2008;12(4):339–45.
[12] Wong KKH, Jankelson D, Reid A, Unger G, Dungan G, Hedner JA, et al. Diagnostic test evaluation of a nasal flow monitor for obstructive sleep apnea detection in sleep apnea research. Behav Res Methods 2008;40(1):360–6.
[13] Hafezi M, Montazeri N, Saha S, Zhu K, Gavrilovic B, Yadollahi A, et al. Sleep apnea severity estimation from tracheal movements using a deep learning model. IEEE Access 2020;8:22641–9.
[14] Saha S, Kabir M, Montazeri N, Hafezi M, Gavrilovic B, Zhu K, et al. Portable diagnosis of sleep apnea with the validation of individual event detection. Sleep Med 2020;; 69::51–-57.
[15] Yadollahi A, Giannouli E, Moussavi Z. Sleep apnea monitoring and diagnosis based on pulse oximetery and tracheal sound signals. Med Biol Eng Comput 2010;48(11):1087–97.
[16] Ballester E, Solans M, Vila X, Hernandez L, Quinto L, Bolivar I, et al. Evaluation of a portable respiratory recording device for detecting apnoeas and hypopnoeas in subjects from a general population. Eur Respir J 2000;16(1):123–7.
[17] Zou D, Grote L, Peker Y, Lindblad U, Hedner J. Validation a portable monitoring device for sleep apnea diagnosis in a population based cohort using synchronized home polysomnography. Sleep 2006;29(3):367–74.
[18] Ng SSS, Chan T, To K, Ngai J, Tung A, Ko FWS, et al. Validation of a portable recording device (ApneaLink) for identifying patients with suspected obstructive sleep apnoea syndrome. Intern Med J 2009;39(11):757–62.
[19] Claman D, Murr A, Trotter K. Clinical validation of the Bedbugg™ in detection of obstructive sleep apnea. Otolaryngology-Head Neck Surg (Tokyo) 2001;125(3):227–30.
[20] Luo Y-M, Tang J, Jolley C, Steier J, Zhong N-S, Moxham J, et al. Distinguishing obstructive from central sleep apnea events: diaphragm electromyogram and esophageal pressure compared. Chest 2009;135(5):1133–41.
[21] Pillar G, Berall M, Berry R, Etzioni T, Shrater N, Hwang D, et al. Detecting central sleep apnea in adult patients using WatchPAT—a multicenter validation study. Sleep Breath 2019:1–12.
[22] Sommermeyer D, Zou D, Grote L, Hedner J. Detection of sleep disordered breathing and its central/obstructive character using nasal cannula and finger pulse oximeter. J Clin Sleep Med 2012;8(5):527–33.
[23] Thomas RJ, Mietus JE, Peng C-K, Gilmartin G, Daly RW, Goldberger AL, et al. Differentiating obstructive from central and complex sleep apnea using an automated electrocardiogram-based method. Sleep 2007;30(12):1756–69.

[24] Akbarian S, Ghahjaverestan NM, Yadollahi A, Taati B. Distinguishing obstructive versus central apneas in infrared video of sleep using deep learning: validation study. J J Med Int Res 2020;22(5):e17252.
[25] Jordan AS, McSharry DG, Malhotra A. Adult obstructive sleep apnoea. Lancet 2014;383(9918):736–47.
[26] Yumino D, Bradley TD. Central sleep apnea and Cheyne-Stokes respiration. Proc Am Thorac Soc 2008;5(2):226–36.
[27] Mooney AM, Abounasr KK, Rapoport DM, Ayappa I. Relative prolongation of inspiratory time predicts high versus low resistance categorization of hypopneas. J Clin Sleep Med 2012;8(2):177–85.
[28] Randerath W, Verbraecken J, Andreas S, Arzt M, Bloch KE, Brack T, et al. Definition, discrimination, diagnosis and treatment of central breathing disturbances during sleep. J European respiratory journal 2017;49(1):1600959.
[29] Saha S, Bradley TD, Taheri M, Moussavi Z, Yadollahi A. A subject-specific acoustic model of the upper airway for snoring sounds generation. Sci Rep 2016;6:25730.
[30] Koo SK, Choi JW, Myung NS, Lee HJ, Kim YJ, Kim YJ. Analysis of obstruction site in obstructive sleep apnea syndrome patients by drug induced sleep endoscopy. Am J Otolaryngol 2013;34(6):626–30.
[31] Won T-B, Kim SY, Lee WH, Han DH, Kim D-Y, Kim J-W, et al. Acoustic characteristics of snoring according to obstruction site determined by sleep videofluoroscopy. Acta Otolaryngol 2012;132(sup1):S13–20.
[32] Rembold CM, Suratt PM. Airway turbulence and changes in upper airway hydraulic diameter can be estimated from the intensity of high frequency inspiratory sounds in sleeping adults. J Physiol 2014;592(Pt 17):3831–9.
[33] Alshaer H, Fernie GR, Tseng W-H, Bradley TD. Comparison of in-laboratory and home diagnosis of sleep apnea using a cordless portable acoustic device. Sleep Med 2016;22:91–6.
[34] Kalkbrenner C, Eichenlaub M, Rüdiger S, Kropf-Sanchen C, Rottbauer W, Brucher R. Apnea and heart rate detection from tracheal body sounds for the diagnosis of sleep-related breathing disorders. Med Biol Eng 2018;56(4):671–81.
[35] Nakano H, Furukawa T, Tanigawa T. Tracheal sound analysis using a deep neural network to detect sleep apnea. J Clin Sleep Med 2019;15(8):1125–33.
[36] Nakano H, Hayashi M, Ohshima E, Nishikata N, Shinohara T. Validation of a new system of tracheal sound analysis for the diagnosis of sleep apnea-hypopnea syndrome. Sleep 2004;27(5):951–7.
[37] Ben-Israel N, Tarasiuk A, Zigel Y. Obstructive apnea hypopnea index estimation by analysis of nocturnal snoring signals in adults. Sleep 2012;35(9):1299. 305C.
[38] Penzel T, Sabil A. The use of tracheal sounds for the diagnosis of sleep apnoea. Breathe 2017;13(2):e37–45.
[39] Ghahjaverestan NM, Akbarian S, Hafezi M, Saha S, Zhu K, Gavrilovic B, et al. Sleep/Wakefulness detection using tracheal sounds and movements. Nat Sci Sleep 2020;12:1009–21.
[40] Ghahjaverestan NM, Kabir M, Saha S, Zhu K, Gavrilovic B, Alshaer H, et al. Automatic respiratory phase identification using tracheal sounds and movements during sleep. Ann Biomed Eng 2021:1–13.
[41] Saha S, Moussavi Z, Hadi P, Bradley TD, Yadollahi A. Effects of increased pharyngeal tissue mass due to fluid accumulation in the neck on the acoustic features of snoring sounds in men. J Clin Sleep Med 2018;14(10):1653–60.
[42] Saha S, Viswanathan K, Saha A, Yadollahi A. Feasibility of acoustic features of vowel sounds in estimating the upper airway cross sectional area during wakefulness: a pilot study. Speech Commun 2024;165:103144.
[43] Saha S, Rattansingh A, Martino R, Viswanathan K, Saha A, Montazeri Ghahjaverestan N, et al. A pilot observation using ultrasonography and vowel articulation to investigate the influence of suspected obstructive sleep apnea on upper airway. Sci Rep 2024;14(1):6144.
[44] Ghahjaverestan NM, Saha S, Kabir M, Gavrilovic B, Zhu K, Yadollahi A. Sleep apnea severity based on estimated tidal volume and snoring features from tracheal signals. J Sleep Res 2022;31(2):e13490.
[45] Lee L-A, Lo Y-L, Yu J-F, Lee G-S, Ni Y-L, Chen N-H, et al. Snoring sounds predict obstruction sites and surgical response in patients with obstructive sleep apnea hypopnea syndrome. Sci Rep 2016;6.
[46] Teerapraipruk B, Chirakalwasan N, Simon R, Hirunwiwatkul P, Jaimchariyatam N, Desudchit T, et al. Clinical and polysomnographic data of positional sleep apnea and its predictors. Sleep brea 2012;16(4):1167–72.
[47] Saha S, Gruenspan G, Kabir MM, Montazeri N, Hafezi M, Meisami T, et al. Investigation of the relationship between snoring sounds features and response to the mandibular advancement devices. Sleep Med 2019;64:S331.
[48] Zhou Z, Shin J, Zhang L, Gurudu S, Gotway M, Liang J. Fine-tuning convolutional neural networks for biomedical image analysis: actively and incrementally. 2017.
[49] Krizhevsky A, Sutskever I, Hinton GE. Imagenet classification with deep convolutional neural networks. 2012.
[50] He K, Zhang X, Ren S, Sun J. Deep residual learning for image recognition. 2016.
[51] Simonyan K, Zisserman A. Very deep convolutional networks for large-scale image recognition. Arxiv Preprint 2014.
[52] Huang G, Liu Z, Van Der Maaten L, Weinberger KQ. Densely connected convolutional networks. 2017.
[53] Paszke A, Gross S, Massa F, Lerer A, Bradbury J, Chanan G, et al. Pytorch: an imperative style, high-performance deep learning library. 2019.
[54] Khayat R, Abraham W, Patt B, Brinkman V, Wannemacher J, Porter K, et al. Central sleep apnea is a predictor of cardiac readmission in hospitalized patients with systolic heart failure. J Card Fail 2012 Jul;18(7):534–40.
[55] Somers VK, White DP, Amin R, Abraham WT, Costa F, Culebras A, et al. Sleep apnea and cardiovascular disease: an American heart association/American college of cardiology foundation scientific statement from the American heart association council for high blood pressure research professional education committee, council on clinical cardiology, stroke council, and council on cardiovascular nursing. J Am Coll Cardiol 2008 Aug 19;52(8):686–717.
[56] Walker JM, Farney RJ, Rhondeau SM, Boyle KM, Valentine K, Cloward TV, et al. Chronic opioid use is a risk factor for the development of central sleep apnea and ataxic breathing. J Clin Sleep Med 2007 Aug 15;3(5):455–61.
[57] Eckert DJ, Jordan AS, Merchia P, Malhotra A. Central sleep apnea: pathophysiology and treatment. Chest 2007 Feb;131(2):595–607.
[58] White DP, Zwillich CW, Pickett CK, Douglas NJ, Findley LJ, Weil JV. Central sleep apnea. Improvement with acetazolamide therapy. Arch Intern Med 1982 Oct;142 (10):1816–9.
[59] Selim BJ, Junna MR, Morgenthaler TI. Therapy for sleep hypoventilation and central apnea syndromes. Curr Treat Options Neurol 2012 Oct;14(5):427–37.
[60] Azarbarzin A, Moussavi Z. Snoring sounds variability as a signature of obstructive sleep apnea. Med Eng Phys 2013 Apr;35(4):479–85.
[61] Maimon N, Hanly PJ. Does snoring intensity correlate with the severity of obstructive sleep apnea? J Clin Sleep Med 2010 Oct 15;6(5):475–8.
[62] Nakano H, Hirayama K, Sadamitsu Y, Toshimitsu A, Fujita H, Shin S, et al. Monitoring sound to quantify snoring and sleep apnea severity using a smartphone: proof of concept. J Clin Sleep Med 2014;10(1):73–8.
[63] Yadollahi A, Montazeri A, Azarbarzin A, Moussavi Z. Respiratory flow-sound relationship during both wakefulness and sleep and its variation in relation to sleep apnea. Ann Biomed Eng 2013;41(3):537–46.
[64] Yadollahi A, Moussavi ZM. A robust method for estimating respiratory flow using tracheal sounds entropy. IEEE Trans Biomed Eng 2006;53(4):662–8.
[65] Ayappa I, Rapoport DM. The upper airway in sleep: physiology of the pharynx. Sleep Med Rev 2003;7(1):9–33.
[66] Huang L, Quinn SJ, Ellis PD, Williams JE. Biomechanics of snoring. Endeavour 1995;19(3):96–100.
[67] Qian S, Chen Dapang %J. IEEE signal processing magazine. Joint time-frequency analysis 1999;16(2):52–67.
[68] Ghafoorian M, Mehrtash A, Kapur T, Karssemeijer N, Marchiori E, Pesteie M, et al. Transfer learning for domain adaptation in mri: application in brain lesion segmentation. Springer; 2017.
[69] Khan S, Islam N, Jan Z, Din IU, Rodrigues JJPC. A novel deep learning based framework for the detection and classification of breast cancer using transfer learning. Pattern Recognit Lett 2019;125:1–6.
[70] Shie C-K, Chuang C-H, Chou C-N, Wu M-H, Chang EY. Transfer representation learning for medical image analysis. IEEE; 2015.
[71] Talo M, Baloglu UB, Yıldırım Ö, {Acharya UR. Application of deep transfer learning for automated brain abnormality classification using MR images. Cogn Syst Res 2019;54:176–88.